Presentacion Práctica 4

Sistemas Informaticos 2

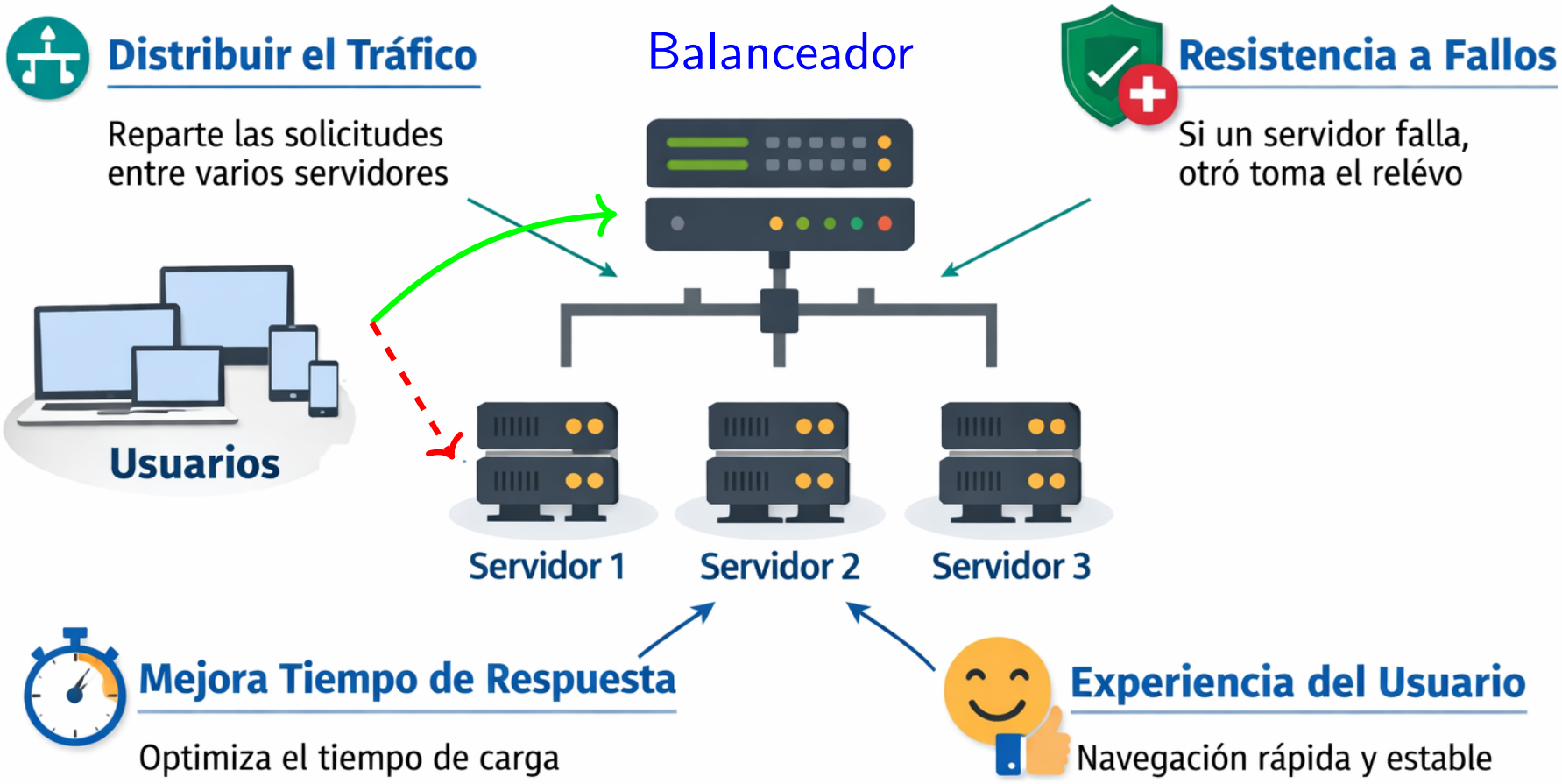
EPS-UAM

2025-26

Objetivos

- Implementar un balanceador de carga
  - Distribuir el tráfico entre varios servidores
- Pruebas de Failover
  - Verificar capacidad de sistema de seguir operando en caso de fallar una instancia con recursos de respaldo.
- Análisis de rendimiento
  - Ejecución de pruebas de rendimiento con JMeter

Balanceador de Carga



Apache Load Balancer: consola

127.0.0.1:18080/balancer-manager 110%

Load Balancer Manager for 127.0.0.1

Server Version: Apache/2.4.58 (Ubuntu)
Server Built: 2024-10-02T12:40:51
Balancer changes will NOT be persisted on restart.
Balancers are inherited from main server.
ProxyPass settings are inherited from main server.

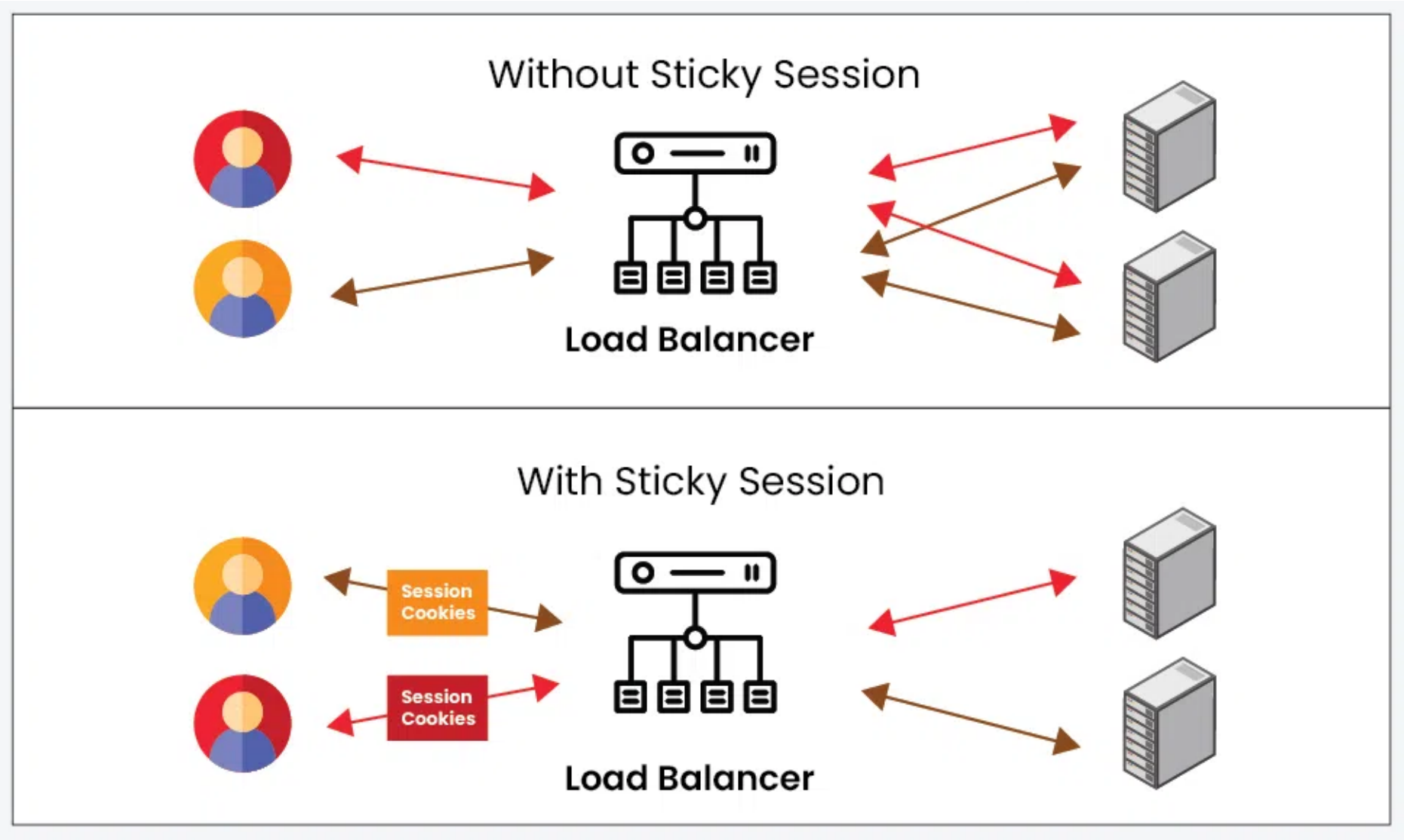
LoadBalancer Status for balancer://si2cluster [p2c258133_si2cluster]

MaxMembers	StickySession	DisableFailover	Timeout	FailoverAttempts	Method	Path	Active
3 [3 Used]	ROUTEID	Off	0	2	byrequests	/P1base	Yes

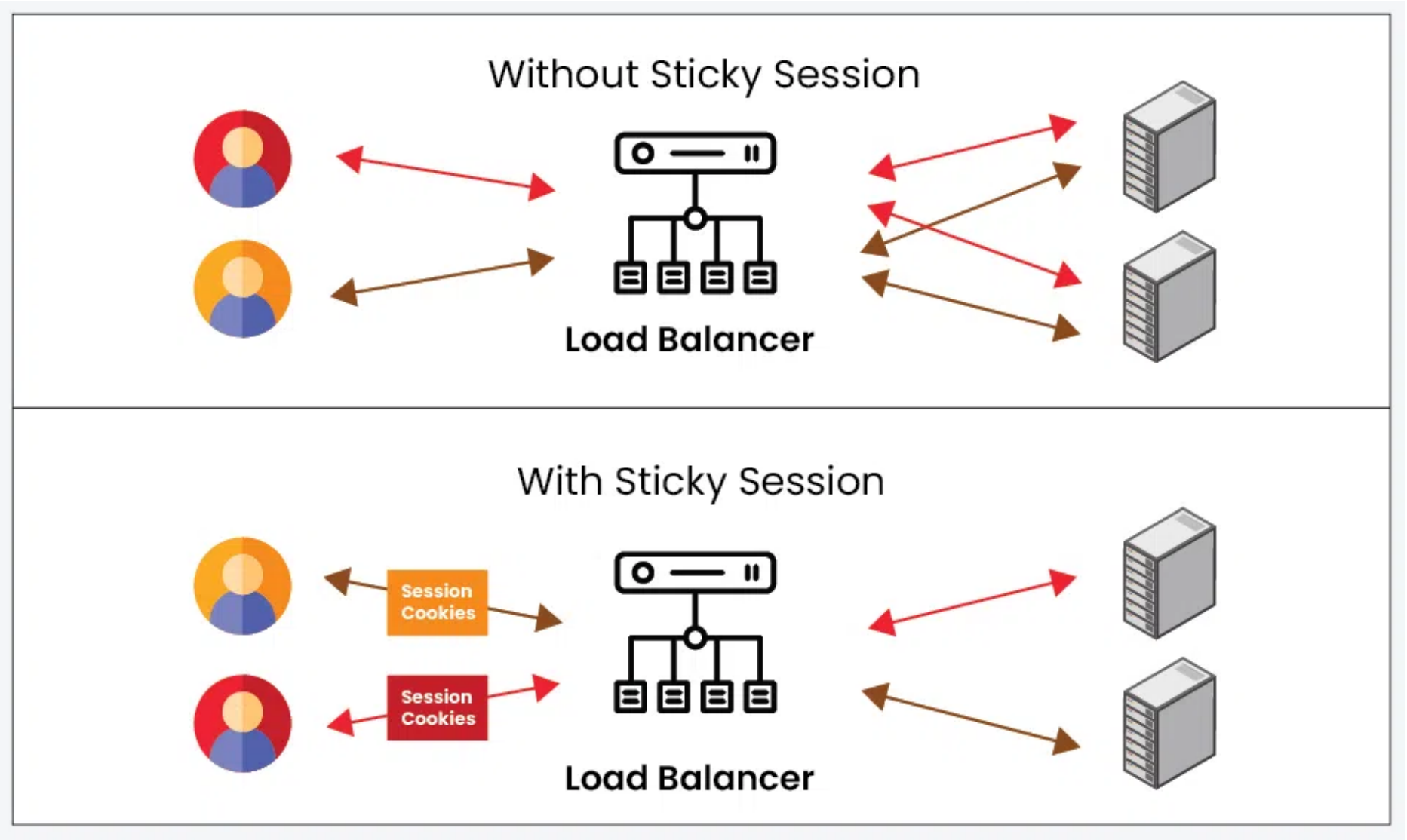
Worker URL	Route	RouteRedir	Factor	Set	Status	Elected	Busy	Load	To	From
http://10.0.2.2:18000	Instance01		1.00	0	Init Ok	0	0	0	0	0
http://10.0.2.2:28000	Instance02		1.00	0	Init Ok	0	0	0	0	0
http://10.0.2.2:38000	Instance03		1.00	0	Init Ok	0	0	0	0	0

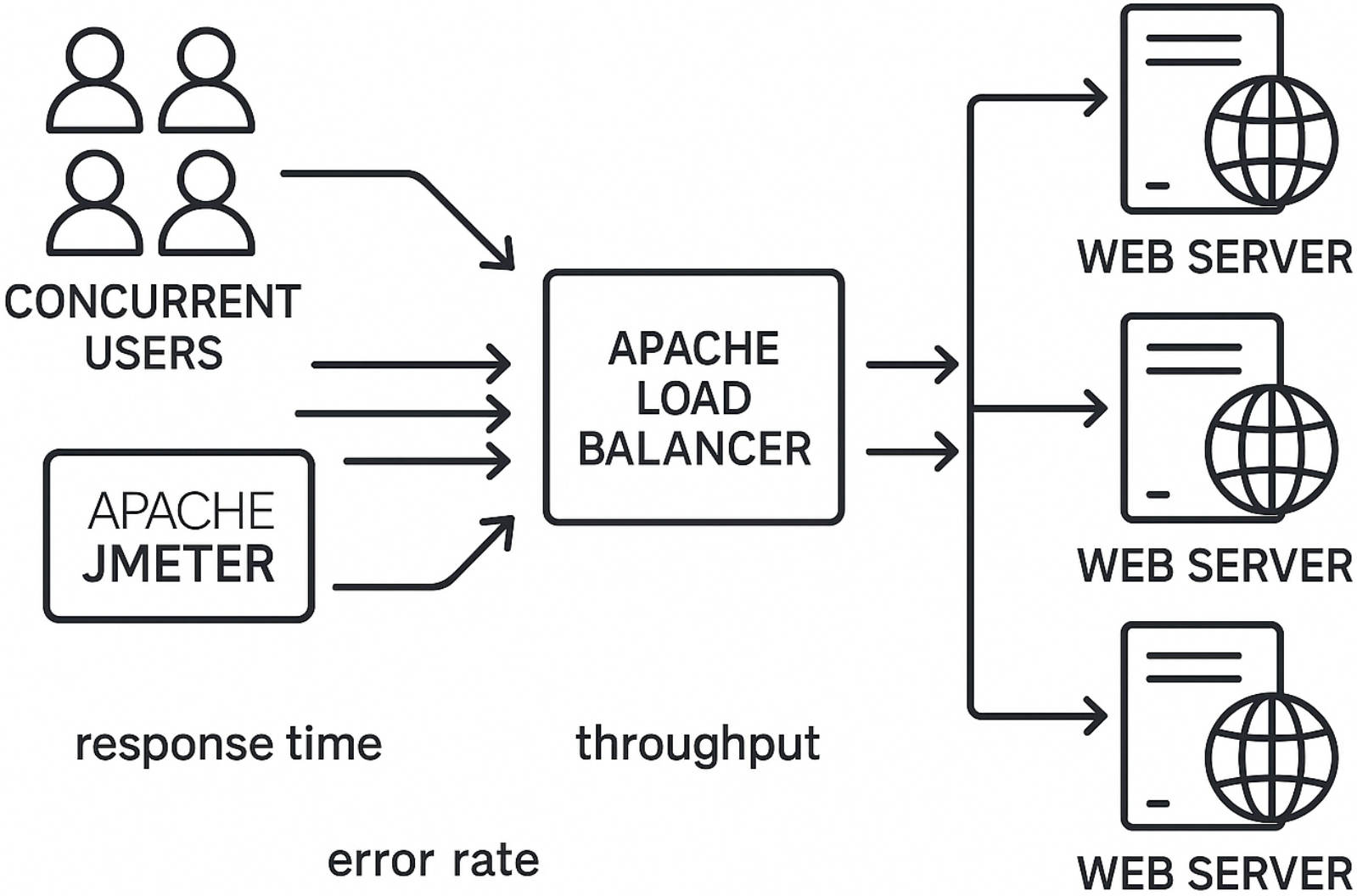
Apache/2.4.58 (Ubuntu) Server at 127.0.0.1 Port 18080

Apache Load Balancer: sticky session



Apache Load Balancer: sticky session



¿Qué ocurre si un servidor falla?

Medir Rendimiento